



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

04-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	
	A.	C	C	C	B	B	
SECTION-I (ii)	Q.	6	7	8	9		
	A.	A,B,C	A,B	A,D	A,D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	16	2	9	8	6	320
SECTION-IV	Q.	1	2				
	A.	A→Q,B→PR,C→S,D→T	A→Q,B→S,C→R,D→P				

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	
	A.	D	D	D	A	A	
SECTION-I (ii)	Q.	6	7	8	9		
	A.	A	A,B,C,D	A,C,D	B,D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	2	5	1	9	5	2
SECTION-IV	Q.	1	2				
	A.	A→PQ,B→PST,C→PR,D→P	A→P,B→PQS,C→PR,D→QST				

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	
	A.	B	D	A	D	A	
SECTION-I (ii)	Q.	6	7	8	9		
	A.	A,B,D	A,C	A,D	A		
SECTION-II	Q.	1	2	3	4	5	6
	A.	140	255	6	5	1365	33
SECTION-IV	Q.	1	2				
	A.	A→P,B→R,C→T,D→R	A→ST,B→ST,C→QRST,D→PQRST				

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

The decay law will remain same even in the train. The velocities of the α -particle and the recoiling nucleus will be same on the ground and in the train with respect to train.

2. Ans (C)

Centre point is the unstable equilibrium position where potential energy is maximum.

3. Ans (C)

Gravitational field intensity due to ring,

$$I_g = \frac{\sqrt{3}GMa}{(a^2 + 3a^2)^{3/2}} = \frac{\sqrt{3}GM}{8a^2}$$

$$|F_{\text{sphere}}| = \frac{\sqrt{3}GMm}{8a^2}$$

$$|F_{\text{ring}}| = \frac{\sqrt{3}GMm}{8a^2}$$

$$a_{\text{ring}} = \frac{\sqrt{3}GM}{8a^2}$$

4. Ans (B)

$$mg = kx_{eq} \Rightarrow 4 \times 80 \times 10 = k \cdot \left(\frac{1.6}{100} \right)$$

$$k = 2 \times 10^5 \text{ N/m}$$

$$T = 2\pi \sqrt{\frac{m}{k}} = \frac{\Delta x}{v}$$

$$2\pi \sqrt{\frac{320}{2 \times 10^5}} = \frac{8}{v} = 2\pi \sqrt{16 \times 10^{-4}}$$

$$\frac{8}{v} = 2\pi \times 4 \times 10^{-2}$$

$$v = \frac{8}{8\pi \times 10^{-2}} = \frac{\pi}{\pi^2} \times 100 = 10\pi$$

5. Ans (B)

Case-I :

$$f_A = f_C - 8 \quad f_B = f_A (1 + 0.06)$$

$$f_B = f_C + 7$$

$$f_B - f_A = 15$$

$$f_A (0.06) = 15$$

$$f_A = 250 \text{ Hz}$$

$$f_B = 265 \text{ Hz}$$

$$f_C = 258 \text{ Hz}$$

Case-II :

$$f_A = f_C + 8 \quad (\text{Not possible})$$

$$f_B = f_C + 7$$

Case-III

$$f_A = f_C - 8$$

$$f_B = f_C - 7$$

$$f_B - f_A = 1$$

$$f_A (1 + 0.06) - f_A = 1$$

$$f_A \times 0.06 = 1$$

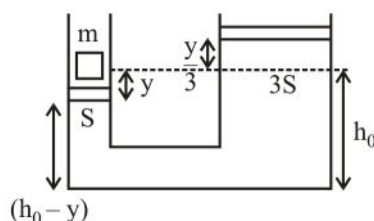
$$f_A = 16.67 \text{ (Not in audible range)}$$

Hence only case-I is possible.

PART-1 : PHYSICS

SECTION-I (ii)

6. Ans (A,B,C)



Assuming that the wires have a resistance per unit length α .

$$V_0 = I_0 (2\alpha h_0 + R_A)$$

$$P_0 + \frac{mg}{S} + \rho g (h_0 - y) = P_0 + \rho g \left(h_0 + \frac{y}{3} \right)$$

$$\frac{mg}{S} = \rho g \left(\frac{y}{3} + y \right) = \rho g \cdot \frac{4y}{3} = \frac{4}{3} \rho g y$$

$$y = \frac{3}{4} \frac{m}{\rho S}$$

$$V_0 = I_1 \left(R_A + \alpha (h_0 - y) + \alpha \left(h_0 + \frac{y}{3} \right) \right)$$

$$= I_1 \left(R_A + 2\alpha h_0 - \frac{2\alpha y}{3} \right)$$

$$= I_1 \left(R_A + 2\alpha h_0 - \frac{2\alpha}{3} \cdot \frac{3}{4} \frac{m}{\rho S} \right)$$

$$V_0 = I_1 \left(R_A + 2\alpha h_0 - \frac{\alpha m}{2\rho S} \right) \dots (i)$$

$$V_0 = I_2 \left(R_A + 2\alpha h_0 + \frac{\alpha m}{2\rho S} \right) \dots (ii)$$

$$2.31 \left(R_A + 2\alpha h_0 - \frac{\alpha m}{2\rho S} \right) = 2.1 \left(R_A + 2\alpha h_0 + \frac{\alpha m}{2\rho S} \right)$$

$$0.21 (R_A + 2\alpha h_0) = (4.41) \frac{\alpha m}{2\rho S}$$

$$R_A + 2\alpha h_0 = 21 \cdot \frac{\alpha m}{2\rho S} \dots (iii)$$

$$V_0 = I_2 \left(\frac{21\alpha m}{2\rho S} + \frac{\alpha m}{2\rho S} \right) = I_2 \cdot \frac{22\alpha m}{2\rho S} = \frac{11\alpha m}{\rho S} \cdot I_2$$

$$46.2 = \frac{11\alpha \times 1}{10^3 \times 100 \times 10^{-4}} \times 2.1$$

$$\alpha = 20$$

$$R_A + 2 \times 20 \times \frac{1}{2} = \frac{21 \times 20 \times 1}{2 \times 10^3 \times 100 \times 10^{-4}}$$

$$R_A + 20 = 21$$

$$R_A = 1 \Omega$$

when no load is present,

$$V = I_0 (2\alpha h_0 + R_A)$$

$$46.2 = I_0 \left(2 \times 20 \times \frac{1}{2} + 1 \right)$$

$$I_0 = 2.2 \text{ A}$$

For maximum current, $Sh_0 + 3Sh_0 = 3Sh_1$

$$h_1 = \frac{4}{3}h_0$$

$$46.2 = I_{\max} \left(R_A + \frac{4}{3}\alpha h_0 \right)$$

$$= I_{\max} \left(1 + \frac{4}{3} \times 20 \times \frac{1}{2} \right) = I_{\max} \left(1 + \frac{4}{3} \times 10 \right)$$

$$I_{\max} = \frac{46.2}{14.33} = 3.22 \text{ A}$$

For minimum current, $Sh_0 + 3Sh_0 = Sh_2$

$$h_2 = 4h_0$$

$$I_{\min} = \frac{46.2}{R_A + \alpha(4h_0)}$$

$$= \frac{46.2}{1 + 20 \times 4 \times \frac{1}{2}} = \frac{46.2}{41} = 1.126 \text{ A}$$

7. Ans (A,B)

Since the angles in a triangle formed by tigers remain constant, at any moment in time the triangle formed by tigers is similar to the original one. It follows from this that all tigers will meet at the same time.

Let's find the rate of change of distance AB. Since the speed modules of the tigers and the angles between them remain constant, the required speed is equal v .

The time before the first and second tigers meet is $t = L/v$.

Let us express the rates of change of distances BC and AC:

$$v_{BC} = v_2 + \frac{v_3}{\sqrt{2}},$$

$$v_{AC} = v_3 + \frac{v_1}{\sqrt{2}}$$

Since the angles in triangle ABC remain constant, the relationships between its sides also remain constant. This means that the rates of change of the sides of the triangle are proportional to them

$$\frac{v_{AB}}{AB} = \frac{v_{BC}}{BC} = \frac{v_{AC}}{AC},$$

$$\frac{v}{L} = \frac{v_3 + \frac{v}{\sqrt{2}}}{L\sqrt{2}} \Rightarrow v_3 = \frac{v}{\sqrt{2}},$$

$$\frac{v}{L} = \frac{v_2 + \frac{v}{2}}{L} \Rightarrow v_2 = \frac{v}{2}.$$

Since the velocity modulus of the second tiger is always half the velocity modulus of the first tiger, and the directions of velocities at any moment are mutually perpendicular, the trajectories of their movement will be similar to each other with a similarity coefficient of 2 : 1. It follows that the displacement vectors will also be mutually perpendicular and differ in magnitude by a factor of two.

This can also be proven strictly mathematically.

Let's consider a small moment of time dt . Let the angle between the speed of the first tiger and the x-axis be equal to α , then the angle between the speed of the second tiger and the x-axis will be equal to $\alpha + 90^\circ$. Let us write down the changes in the coordinates of tigers in a short time:

$$dx_1 = v \cos \alpha \cdot dt,$$

$$dy_1 = v \sin \alpha \cdot dt,$$

$$dx_2 = \frac{v}{2} \cos(\alpha + 90^\circ) \cdot dt = -\frac{v}{2} \sin \alpha \cdot dt,$$

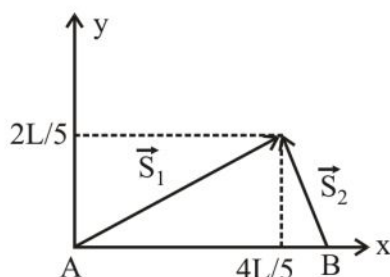
$$dy_2 = \frac{v}{2} \sin(\alpha + 90^\circ) = \frac{v}{2} \cos \alpha \cdot dt.$$

From the resulting equations it is clear that for any moment of time $dx_1 = 2dy_2$, $dy_1 = -2dx_2$, this means that the vectors of total displacements also differ by a factor of two in absolute value and are mutually perpendicular

$$\text{Since the tigers met: } \vec{S}_1 = \vec{AB} + \vec{S}_2.$$

Then from the right triangle we find

$$(x_B, y_B) = \left(\frac{4L}{5}, \frac{2L}{5} \right).$$



8. Ans (A,D)

According to Fermat's principle, all rays of light incident as parallel light and converging at point F should have equal optical path lengths, that is, the optical path lengths of the edgemost light BF and any light NMF should be equal [M is any point on the convex surface, its coordinates is $M(x,y)$], from this, the equation of the convex surface can be determined, and other problems can also be easily solved.

Taking the Oxy coordinate as shown in Figure 1, based on the isopticality of light BF and NMF, we get

$$L = nx + \sqrt{(f-x)^2 + y^2} = \sqrt{f^2 + R^2}$$

Squaring the above formula and sorting it out a little, we can get the equation that the coordinates (x, y) of any point M (x, y) on the convex surface of the lens should satisfy

$$(n^2 - 1) \left(x - \frac{n\sqrt{f^2 + R^2} - f}{n^2 - 1} \right)^2 - y^2 = \frac{(nf - \sqrt{f^2 + R^2})^2}{n^2 - 1}$$

$$x_0 = \frac{n\sqrt{f^2 + R^2} - f}{n^2 - 1}$$

$$a = \frac{nf - \sqrt{f^2 + R^2}}{\sqrt{n^2 - 1}}$$

$$(n^2 - 1)(x - x_0)^2 - y^2 = a^2$$

This is the equation of the hyperbola. Due to rotational symmetry, the convex surface of the lens should be a rotational hyperboloid.

The position X_A of the lens vertex A should satisfy

$$(n^2 - 1)(x_A - x_0)^2 = a^2$$

$$x_A = x_0 + \frac{a}{\sqrt{n^2 - 1}} = \frac{\sqrt{f^2 + R^2} + f}{n + 1}$$

Since point F is outside the lens, that is $x_A \leq f$

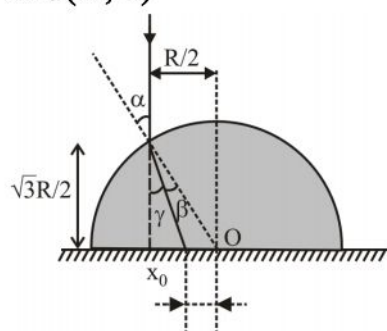
$$\text{This is a restriction on R, that is } \frac{\sqrt{f^2 + R^2} + f}{n + 1} \leq f$$

$$R \leq \sqrt{n^2 - 1} f$$

In the extreme case, that is, when

$$R = \sqrt{n^2 - 1} f$$

9. Ans (A,D)



(a) Let's find the point on the table where the beam hits at the very first moment $t = 0$. Consider Figure, it is easy to find that the angle of incidence of the beam on the hemisphere is $\alpha = 30^\circ$, i.e. $\sin \alpha = \frac{1}{2}$.

According to Snell's law of refraction

$$\sin \beta = \frac{\sin \alpha}{n} = \frac{1}{2n} \Rightarrow \tan \beta = \frac{1}{\sqrt{4n^2 - 1}}$$

From the triangle with angle γ in Figure it is easy to find its horizontal leg $\frac{\sqrt{3}R}{2} \tan \gamma$, which means the point of incidence of the beam on the table: it is located at a distance x_0 from the center of hemisphere O:

$$x_0 = \frac{R}{2} - \frac{\sqrt{3}}{2} R \tan \gamma$$

$$\tan \gamma = \tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} = \frac{\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{4n^2 - 1}}}{1 + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{4n^2 - 1}}}$$

$$x_0 = \frac{R}{2} \left(1 - \frac{\sqrt{12n^2 - 3} - 3}{\sqrt{12n^2 - 3} + 1} \right) = \frac{2R}{\sqrt{12n^2 - 3} + 1}$$

(b) At time t , the base in contact with the table is a circle with radius r . Within a time interval dt , the amount of heat absorbed is

$$dQ = K\pi r^2 dt$$

for some constant K . The volume of the ice melt is

$$dQ = Ldm = L\rho\pi r^2 dz$$

Where L and ρ are the latent heat and density of the ice respectively. Therefore we have

$$K\pi r^2 dt = -L\rho\pi r^2 dz$$

$$\Rightarrow \frac{dz}{dt} = -\frac{K}{L\rho} \text{ is a constant}$$

i.e. the melting uniformly reduces the thickness of the ice, we have

$$z(t) = R \left(1 - \frac{t}{T_0} \right)$$

(c) At the end, when the base of the ice has radius $r \leq \frac{R}{2}$, the beam will obviously hit the table vertically (i.e. $x(t) = \frac{R}{2}$). It corresponds to the case

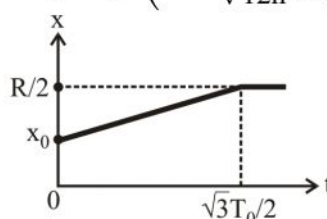
when the thickness of the ice reduces to $z(t)$

$$= \left(1 - \frac{\sqrt{3}}{2} \right) R. \text{ It will happen at the time } t = \frac{\sqrt{3}}{2} T_0. \text{ Therefore, for } t > \frac{\sqrt{3}}{2} T_0,$$

$$\text{we have } x(t) = \frac{R}{2}$$

To get $x(t)$ for $t < \frac{\sqrt{3}}{2} T_0$, we can consider the similar triangles.

$$\begin{aligned} \frac{\frac{R}{2} - x_0}{\frac{R}{2} - x(t)} &= \frac{\frac{\sqrt{3}}{2} R}{R \left(\frac{\sqrt{3}}{2} - \frac{t}{T_0} \right)} \\ \Rightarrow \left(\frac{R}{2} - x_0 \right) \left(\frac{\sqrt{3}}{2} - \frac{t}{T_0} \right) &= \frac{\sqrt{3}}{2} \left(\frac{R}{2} - x \right) \\ \Rightarrow x(t) &= \frac{R}{2} - \left(\frac{R}{2} - x_0 \right) \left(1 - \frac{2}{\sqrt{3}} \frac{t}{T_0} \right) \\ &= \frac{R}{2} - \frac{R}{2} \left(1 - \frac{4}{\sqrt{12n^2 - 3} + 1} \right) \left(1 - \frac{2}{\sqrt{3}} \frac{t}{T_0} \right) \end{aligned}$$

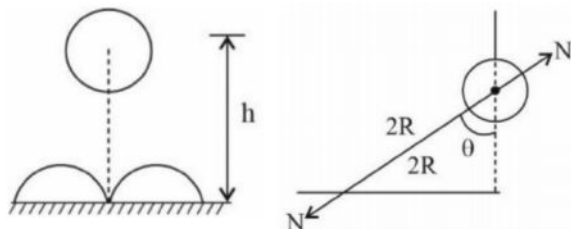


PART-1 : PHYSICS

SECTION-II

3. Ans (9)

$$\sin \theta = \frac{1}{\sqrt{3}}$$



$$\therefore \cos \theta = \sqrt{\frac{2}{3}} \therefore \int 3N \cos \theta dt = mv_0$$

$$\int N \sin \theta dt = \frac{m}{2} v$$

$$\therefore 3 \cot \theta = \frac{2v_0}{v}$$

$$\therefore v = \frac{2}{3\sqrt{2}} v_0 = \frac{\sqrt{2}}{3} v_0 \therefore p = \frac{\frac{1}{2} m v_0^2}{\frac{1}{2} \cdot \frac{m}{2} \cdot \frac{2v_0^2}{9}} = 9$$

4. Ans (8)

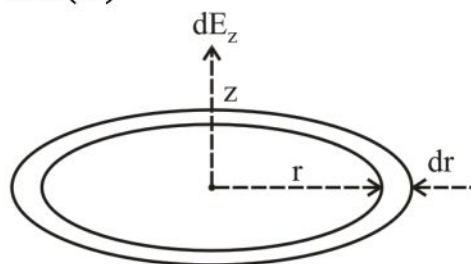
$$\text{Height of water column} = \frac{20}{5} = 4\text{cm}$$

$$\text{Mass of water} = 20 \text{ gm}$$

By work energy theorem

$$N^{\text{ext}} = \Delta U^{\text{Piston}} + \Delta U^{\text{Water}} + KE^{\text{Water}} = 8 \times 10^{-2} \text{ J}$$

5. Ans (6)



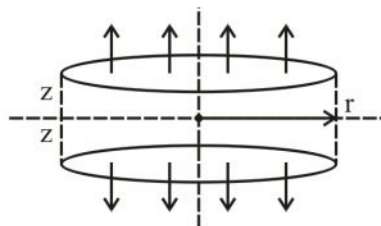
for $z \ll r$

$$dq = \sigma(2\pi r) dr$$

$$E_z = k\sigma(2\pi)z \int_{r_1}^{r_2} \frac{dr}{r^2} \\ = k\sigma(2\pi)z \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

In order to calculate radial component of $\vec{E}$ in

the plane of ring for $r < R_1$



$$E_z(2\pi r^2) + E_r(2\pi(2z)r) = 0$$

$$E_z(2\pi r^2) = -E_r(4\pi rz)$$

$$E_r = -\frac{E_z r}{2z}$$

$$= -\frac{r}{2z} \cdot k\sigma(2\pi)z \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$E_r = -\frac{\sigma}{4\epsilon_0} \cdot r \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

at $r = R$,

$$\frac{kQ}{R^2} = \frac{\sigma R}{4\epsilon_0} \cdot R \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

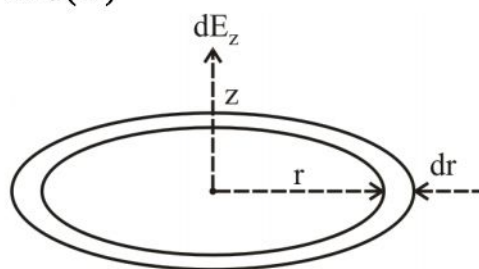
$$\frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R^2} = \frac{\sigma R^2}{4\epsilon_0} \cdot \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

$$\sigma = \frac{Q}{\pi R^3} \left(\frac{r_1 r_2}{r_2 - r_1} \right)$$

$$\text{for } R = 1\text{cm}, r_1 = 2\text{m}_1, r_2 = 3\text{m}, Q = \pi \times 10^{-6} \text{ C}$$

$$\sigma = 6\text{C/m}^2$$

5. Ans (6)



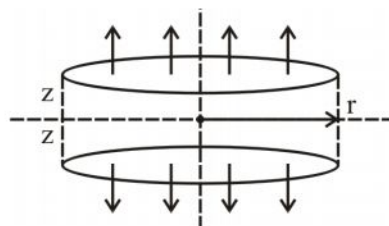
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In order to calculate radial component of $\vec{E}$ in the plane of ring for $r < R_1$



$$E_z(2\pi r^2) + E_r(2\pi(2z)r) = 0$$

$$E_z(2\pi r^2) = -E_r(4\pi rz)$$

$$E_r = -\frac{E_z r}{2z}$$

$$= -\frac{r}{2z} \cdot k\sigma(2\pi)z \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$E_r = -\frac{\sigma}{4\epsilon_0} \cdot r \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

at $r = R$,

$$\frac{kQ}{R^2} = \frac{\sigma R}{4\epsilon_0} \cdot R \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

$$\frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R^2} = \frac{\sigma R^2}{4\epsilon_0} \cdot \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

$$\sigma = \frac{Q}{\pi R^3} \left(\frac{r_1 r_2}{r_2 - r_1} \right)$$

for $R = 1\text{cm}$, $r_1 = 2\text{m}$, $r_2 = 3\text{m}$, $Q = \pi \times 10^{-6}\text{C}$

$$\sigma = 6\text{C/m}^2$$

6. Ans (320)

$$\frac{mgR - \frac{mgR^2}{R+h} - mgh}{mgR - \frac{mgR^2}{R+h}} = \frac{5}{100}$$

$$\frac{Rh - Rh - h^2}{Rh} = \frac{5}{100}$$

$$\frac{h}{R} = \frac{5}{100}$$

$$k = \frac{5}{100} \times 6400 = 320\text{km}$$

PART-1 : PHYSICS

SECTION-IV

1. Ans (A->Q,B->PR,C->S,D->T)

Suppose the plank is displaced by x and at that

instant its speed is v then speed of COM of

cylinder $v/2$. The total energy is

$$E_T = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 + \frac{1}{2}(2m)\left(\frac{v}{2}\right)^2 + \frac{1}{2} \cdot \frac{2mR^2}{2} \left(\frac{v/2}{R}\right)^2$$

$$= \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) mv^2 + \frac{1}{2}kx^2$$

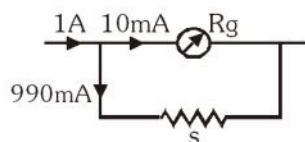
$$= \frac{4+2+1}{8}mv^2 + \frac{1}{2}kx^2 \text{ or } E_T = \frac{7}{8}mv^2 + \frac{1}{2}kx^2$$

$$\text{Also, } E_T = \frac{1}{2}kx_0^2$$

$$\frac{dE_T}{dt} = 0, \text{ after calculating we get } a = -\frac{4k}{7m}x$$

$$\text{Comparing with } a = -\omega^2 x, \text{ we get } \omega = \sqrt{\frac{4k}{7m}}$$

2. Ans (A → Q, B → S, C → R, D → P)



For (A)

$$\vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k} = \frac{1}{x^2}\hat{i} + \frac{1}{y^2}\hat{j} + \frac{1}{z^2}\hat{k}$$

$$\Rightarrow |\vec{E}| = |\hat{i} + \hat{j} + \hat{k}| = \sqrt{3}$$

For (B) $I = \frac{dQ}{dt} = 4 - 4t$, current will be zero

at $t = 1$ s

Total heat produced

$$\int_0^1 I^2 R dt = \int_0^1 (4 - 4t)^2 (6) dt = (16 \times 6) \left[\frac{(1-t)^2}{-3} \right]_0^1 = 32$$

For (C) $(10 \text{ mA}) (R_g) = (990 \text{ mA}) (s)$

$$\Rightarrow s = \frac{0.99}{99} = 0.01 \Omega$$

For (D) $U = -\vec{p} \cdot \vec{E}$ Here $\vec{p} = \pm q d \hat{i}$

$$= \pm (10 \times 10^{-3}) (8 \times 10^{-2}) = \pm (8 \times 10^{-4})$$

$$\Rightarrow U = \pm (8 \times 10^{-4}) (500) = \pm 0.4$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (D)

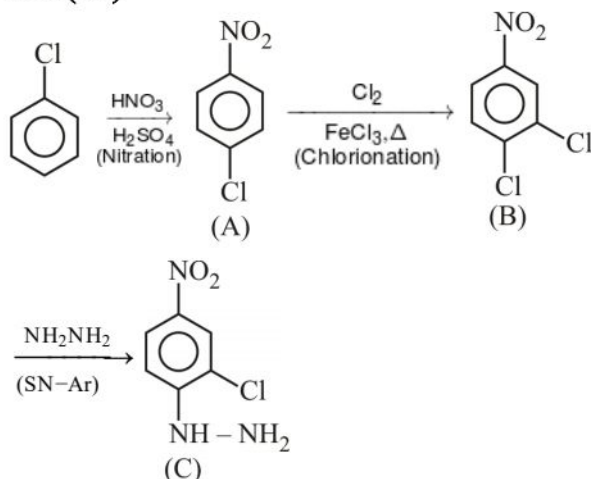
$$\frac{r_A}{r_B} = \frac{2}{1} = \sqrt{\frac{M_B}{M_A}}$$

$$\therefore \frac{M_B}{M_A} = \frac{4}{1}$$

$$\frac{(U_{rms})_A}{(U_{rms})_B} = \sqrt{\frac{T_A}{M_A} \times \frac{M_B}{T_B}} = \sqrt{\frac{2}{1} \times \frac{4}{1}}$$

$$= \frac{2\sqrt{2}}{1}$$

5. Ans (A)



PART-2 : CHEMISTRY

SECTION-I (ii)

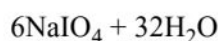
8. Ans (A, C, D)

Gibb's free energy of formation of all the elements in their natural standard state is arbitrarily assumed to be zero.

PART-2 : CHEMISTRY

SECTION-II

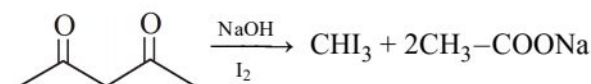
2. Ans (5)



$$-\frac{1}{10} \frac{d[\text{NaOH}]}{dt} = -\frac{1}{2} \frac{d[\text{CrI}_3]}{dt}$$

$$\Rightarrow \frac{\frac{-d[\text{NaOH}]}{dt}}{\frac{-d[\text{CrI}_3]}{dt}} = \frac{10}{2} = \boxed{5}$$

3. Ans (1)



5. Ans (5)

i, iii, vi, vii, viii

PART-2 : CHEMISTRY

SECTION-IV

1. Ans (A → PQ, B → PST, C → PR, D → P)

Crystal system	Crystallographic axes	Crystallographic angles	Possible unit cells
Cubic (Most symmetrical)	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	3 SC, BCC, ECC
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	S, BC 2
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	S, BC, FC, EC 4
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ$ $\beta \neq 90^\circ$ $\beta > 90^\circ$	S, EC 2
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ$ $\gamma = 120^\circ$	S 1
Rhombohedral (Trigonal)	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$ $90^\circ \neq \alpha = \beta = \gamma < 120^\circ$	S 1
Triclinic (Most unsymmetrical)	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	S 1
		Total	14

2. Ans (A → P, B → PQS, C → PR, D → QST)

(A) When $r = 0$, $\psi(r) \neq 0 \Rightarrow$ s-orbital

3 radial node $\Rightarrow$ P

(B) 3 radial node $\Rightarrow$ P, Q, S

(C) For s-orbitals, $\psi(\theta, \phi)$ is independent of θ and ϕ .

(D) Angular node = $\ell \Rightarrow$ Q, S

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$\lim_{n \rightarrow \infty} \sum_{n=1}^n \left(\frac{1}{n^2 \tan^{-1} n^2} - \frac{1}{(n+1)^2 \tan^{-1} (n+1)^2} \right) = \frac{4}{\pi}$$

2. Ans (D)

$$A(B + C) = KI$$

$$B + C = KA^{-1} \quad \dots(1)$$

$$B^T + C^T = K(A^{-1})^T$$

$$B - C = KA \quad \dots(2)$$

$$2B = K(A^{-1} + A)$$

$$2\text{tr}(B) = K \cdot 2 \text{tr}(A) \Rightarrow K = 8$$

3. Ans (A)

$$\frac{1}{7} \times \frac{1}{4} \cdot \left(\frac{3}{6} + \frac{2}{6} + \frac{2}{6} + \frac{2}{6} \right) + \frac{1}{4} \cdot \frac{1}{7} \left(\frac{3}{6} + \frac{2}{6} + \frac{2}{6} + \frac{2}{6} + \frac{2}{6} + \frac{2}{6} + \frac{2}{6} \right)$$

$$\frac{1}{7} \times \frac{1}{4} \times \left(\frac{9}{6} + \frac{15}{6} \right) = \frac{1}{7}$$

4. Ans (D)

$$\vec{v}_1 \cdot \vec{v}_2 = 0 \Rightarrow a = -1, 1$$

$$a = -1; \vec{v}_1 = \hat{i} - 2\hat{j}, \vec{v}_2 = 2\hat{i} + \hat{j}$$

$$a = 1; \vec{v}_1 = \hat{i} - 2\hat{j} + 2\hat{k}, \vec{v}_2 = 2\hat{i} + \hat{j}$$

$$\vec{v}_2 \text{ lies in the plane } 3x - \lambda y + z = 1$$

$$\Rightarrow \lambda = 6$$

Now, volume of the parallelepiped by $\vec{v}_1$, $\vec{v}_2$ and $\vec{n}$

is

$$\begin{vmatrix} 1 & -2 & 2 \\ 2 & 1 & 0 \\ 3 & -6 & 1 \end{vmatrix} = |-25| = 25$$

5. Ans (A)

$$x^2 - p^2 = x^2 - 4x + p^2 \Rightarrow x = \frac{p^2}{2}$$

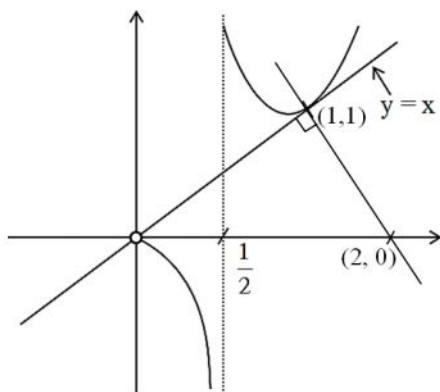
$$\frac{p^4}{4} - p^2 < 0 \Rightarrow p^2(p^2 - 4) < 0 \Rightarrow p \in (-2, 2), p \neq 0$$

$$\therefore |B| = 2$$

PART-3 : MATHEMATICS

SECTION-I (ii)

6. Ans (A,B,D)



$$\frac{d(x^3/y)}{1+x^3/y} = \frac{dx}{x} \Rightarrow \ln(1+x^3/y) = \ln x + \ln c$$

$$1 + \frac{x^3}{y} = cx$$

$$(1,1) \Rightarrow c = 2 \Rightarrow 1 + \frac{x^3}{y} = 2x \Rightarrow y = \frac{x^3}{2x-1}$$

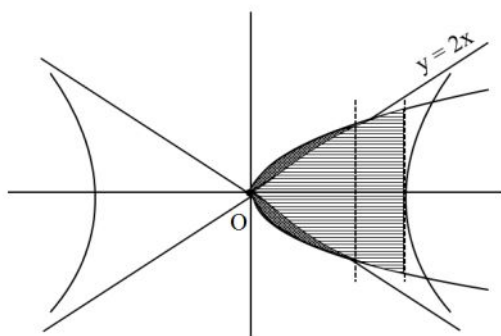
Tangent at (1,1)

$$y - 1 = 1 \cdot (x - 1) \Rightarrow y = x$$

Normal at (1, 1)

$$y - 1 = -(x - 1) \Rightarrow x + y = 2$$

7. Ans (A,C)



For options A and B required area is $\frac{2}{3}$

For options C and D required area is

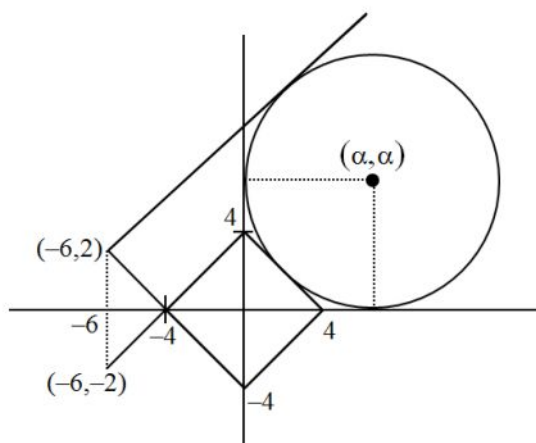
$$\left(\frac{16\sqrt{2}}{3} - \frac{2}{3} \right) \text{ sq. units}$$

8. Ans (A,D)

$$g(x) = \begin{cases} (x^2 - 4)^2 - 4 & x < -\sqrt{6} \text{ or } x > \sqrt{6} \\ 6 - x^2 & -\sqrt{6} \leq x \leq -2 \text{ or } 2 \leq x \leq \sqrt{6} \\ -x & -2 < x < 0 \\ x & 0 \leq x < 2 \end{cases}$$

9. Ans (A)

Circle is -



$$(x - \alpha)^2 + (y - \alpha)^2 = \alpha^2$$

$$\left| \frac{\alpha + \alpha - 4}{\sqrt{2}} \right| = \alpha$$

$$\Rightarrow 2\alpha - 4 = \sqrt{2}\alpha$$

$$\alpha(2 - \sqrt{2}) = 4$$

$$\alpha = \frac{4(2 + \sqrt{2})}{2} = 4 + 2\sqrt{2}$$

maximum length of the tangent will be -

$$= \sqrt{(-6 - \alpha)^2 + (-2 - \alpha)^2 - \alpha^2}$$

$$= \sqrt{\alpha^2 + 16\alpha + 40}$$

$$= \sqrt{(\alpha + 8)^2 - 24} = \sqrt{(12 + 2\sqrt{2})^2 - 24}$$

$$= 4\sqrt{8 + 3\sqrt{2}}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (140)

$$\begin{aligned} & \int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} \ln|3 \cos(x + \frac{\pi}{3}) + 5 \cos x| dx \\ &= \int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} \ln|7 \sin(x + \alpha)| dx \\ &= \pi \ln 7 + \int_0^{\pi} \ln|\sin x| dx \\ &= \pi \ln 7 + 2 \left(-\frac{\pi}{2} \ln 2 \right) \\ &= \pi \ln \frac{7}{2} \\ & \int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} |3 \sin^2 x - 6 \sin x \cos x + \cos^2 x - 2| dx \\ &= \int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} |\cos 2x + 3 \sin 2x| dx = 2\sqrt{10} = \sqrt{40} \\ &\Rightarrow p \cdot q = 140 \end{aligned}$$

2. Ans (255)

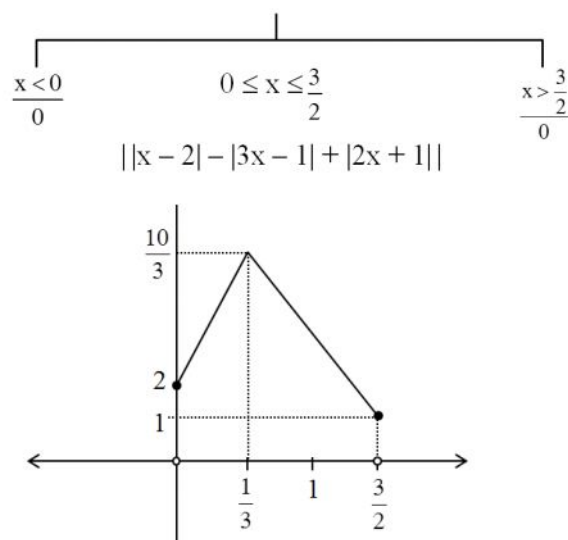
$$\begin{aligned} g(x^3 - 1) &= x^5 + x - 2 \\ \Rightarrow g(t) &= (t+1)^{\frac{5}{3}} + (t+1)^{\frac{1}{3}} - 2 \\ g'(t) &= \frac{5}{3}(t+1)^{\frac{2}{3}} + \frac{1}{3}(t+1)^{-\frac{2}{3}} \\ g(7) &= 32, \quad g'(7) = \frac{27}{4} \\ \frac{d}{dx}((x^3+1)g(x^3)) &= (x^3+1)g'(x^3)3x^2 + g(x^3) \cdot 3x^2 \\ \frac{d}{dx}((x^3+1)g(x^3))|_{x=7^{1/3}} &= \ell = 3 \cdot 7^{2/3} \cdot 86 \\ \Rightarrow \ell(7^{-2/3}) &= 258 \\ \frac{d}{dx} \left(\frac{f(x)}{x^2} \right) &= \frac{x^2 f'(x) - f(x) \cdot 2x}{x^4} \quad f(-2) = -1 \\ &= \frac{xf'(x) - 2f(x)}{x^3} \quad f'(-2) = 0 \\ &\quad f''(-2) = 0 \\ \frac{d^2}{dx^2} \left(\frac{f(x)}{x^2} \right) &= \frac{x^3(xf''(x) + f'(x) - 2f'(x)) - (xf'(x) - 2f(x))3x^2}{x^6} \\ f(-2) &= -1, \quad f'(-2) = 0, \quad f''(-2) = 0 \\ \frac{d^2}{dx^2} \left(\frac{f(x)}{x^2} \right) &= m = \frac{-6}{16} = \frac{-3}{8} \Rightarrow 8m = -3 \\ \therefore \ell(7^{-2/3}) + 8m &= 255 \end{aligned}$$

3. Ans (6)

$$\begin{aligned} \alpha^7 &= 3 \\ x^7 - 21[x^2(x+1) + x^2 + x + 1] + 15 \\ &= 3(1+\alpha)^7 - 21[(\alpha^4 + 2\alpha^3 + \alpha^2)(\alpha^2 + \alpha + 1) \\ &\quad + \alpha^4 + 2\alpha^3 + \alpha^2 + \alpha^2 + \alpha + 1] + 15 \\ &= 6 \end{aligned}$$

4. Ans (5)

$$\begin{aligned} h(x) &= ||x-2| - |3x-1| + |2x+1| | \cdot \left[-x^2 + \frac{3}{2}x + 1 \right] \end{aligned}$$



5. Ans (1365)

$$x + y + z + w = 4 + 8(\sin\theta - \cos\theta)$$

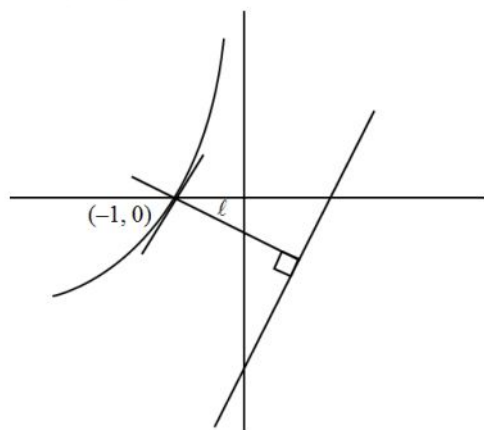
$$4 \leq x + y + z + w \leq \underbrace{4 + 8\sqrt{2}}_{15.3}$$

∴ Natural solutions corresponding to

$$x + y + z + w \leq 15$$

$$= {}^{15}C_4 = 1365$$

6. Ans (33)



Regarding locus will be tangent to the parabola

at $(1/2, -9/8)$ which is $y = 2x - 17/8$

$$y = x - \frac{1}{x}$$

$$y' = 1 + \frac{1}{x^2} = 2 \Rightarrow x = \pm 1$$

$$\therefore \ell = \left| \frac{-2 - \frac{17}{8}}{\sqrt{5}} \right| = \frac{33}{8\sqrt{5}}$$

$$\therefore 8\sqrt{5}\ell = 33$$

PART-3 : MATHEMATICS

SECTION-IV

1. Ans (A → P, B → R, C → T, D → R)

$$(A) -7x + y + 15 = 0$$

$$x - y + 3 = 0$$

$$a_1a_2 + b_1b_2 = -ve$$

∴ Bisector with positive sign will be acute angle bisector as well as origin containing bisector

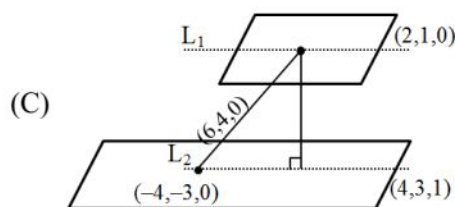
$$\therefore (-7\alpha < +\alpha^2 - 3 + 15)(\alpha - (\alpha^2 - 3) + 3) > 0$$

$$(\alpha^2 - 7\alpha + 12)(\alpha^2 - \alpha - 6) < 0$$

$$(\alpha - 3)(\alpha - 4)(\alpha - 3)(\alpha + 2) < 0$$

$$\alpha \in (-2, 4), \alpha \neq 3$$

$$(B) n((A \times B) \cap (B \times C)) = 4$$



$$\text{dr's of } L_1 \text{ and } L_2 \begin{vmatrix} i & j & k \\ 2 & -3 & 1 \\ 1 & -1 & -1 \end{vmatrix}$$

$$i(4) - j(-3) + k(1)$$

$$4\hat{i} + 3\hat{j} + \hat{k}$$

$$\text{A point on } L_1 : 2x - 3y = 1$$

$$3(x - y = 1)$$

$$-x = -2 \Rightarrow x = 2$$

$$y = 1$$

$$\text{A point on } L_2 : 2x - 3y = 1$$

$$3(x - y = -1)$$

$$-x = 4 \Rightarrow x = -4$$

$$y = -3$$

Projection of $6\hat{i} + 4\hat{j}$ on $4\hat{i} + 3\hat{j} + \hat{k}$ is

$$\frac{24 + 12}{\sqrt{16 + 9 + 1}} = \frac{36}{\sqrt{26}}$$

$$\therefore SD = \sqrt{52 - \frac{36 \times 36}{26}} = 2\sqrt{\frac{7}{13}}$$

$$(D) (i) 4F \quad M_1M_2M_3 \rightarrow 4! \cdot 4!$$

$$(ii) 2F \quad 2F \quad M_1M_2M_3 \rightarrow |M_1| |M_2| |M_3|$$

$$\Rightarrow \frac{4!}{2! \cdot 2! \cdot 2!} \times {}^4C_2 \times 2! \cdot 2! \cdot 2! \times 3!$$

$$= 36(4!)$$

$$\therefore 60(4!)$$

2. Ans (A->ST,B->ST,C->QRST,D->PQRST)

(A) $x^3 + (221 - 4m)x^2 + 220x + 4m$

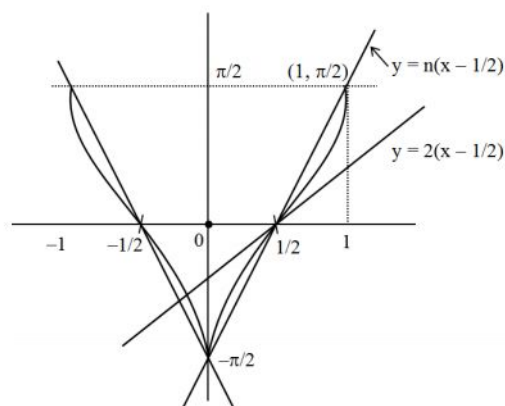
$$= (x + 1)(x^2 - (4m - 220)x + 4m)$$

$$\widehat{\alpha \beta}$$

$$\left. \begin{array}{l} \alpha + \beta = 4m - 220 \\ \alpha\beta = 4m \end{array} \right\} \alpha = 14, \beta = 18$$

$$\frac{\alpha + \beta + m}{19} = \frac{14 + 18 + 63}{19} = 5$$

(B)



Clearly p must lie in $(2, \pi)$

(C)

x_i	1	2	3	n
w_i	1^2	2^2	3^2	n^2
$x_i w_i$	1^3	2^3	3^3	n^3

$$\bar{x}_w = \frac{\sum x_i w_i}{\sum w_i} = \frac{\sum n^3}{\sum n^2} = \frac{\frac{n^2(n+1)^2}{4}}{\frac{n(n+1)(2n+1)}{6}} = \frac{3n(n+1)}{2(2n+1)}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{3n(n+1)}{n \cdot 2(2n+1)} = \frac{3}{4}$$

(D) $f(x) = -x + 7 = f^{-1}(x)$

$$\therefore f(x) - f^{-1}(x) = 0$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

04-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7
	A.	A,B,C	A,B,C,D	A,C	A,C	A,B	C,D	B,C,D
SECTION-I (ii)	Q.	8	9	10	11			
	A.	A	C	D	D			
SECTION-II	Q.	1	2	3	4	5	6	
	A.	21.00	22.50	5.00	40.35 to 40.45	2.00	9.60 to 9.70	

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7
	A.	A,B,D	B,C	A,B	B,C	C,D	B,C	B,C
SECTION-I (ii)	Q.	8	9	10	11			
	A.	D	B	A	B			
SECTION-II	Q.	1	2	3	4	5	6	
	A.	4.00	13.66 to 13.67	9.00	5.00	-530.00	20.00	

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7
	A.	B,C	A,D	A,B,C	A,B,C	A,C	A,B,D	A,B,C
SECTION-I (ii)	Q.	8	9	10	11			
	A.	A	B	B	A			
SECTION-II	Q.	1	2	3	4	5	6	
	A.	1.75	1.51	2.82	1.80	5.50	2.00	

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C)

$$i_1 - i_2 = i_3; i_1 + i_2 = \frac{E}{R}; i_2 = \frac{E}{2R} - \frac{i_3}{2}$$

$$L \frac{di_3}{dt} + i_3 \cdot 2R = i_2 R$$

$$L \frac{di_3}{dt} = R \left(\frac{E}{2R} - \frac{i_3}{2} - 2i_3 \right) i_1$$

$$\int_0^{i_3} \frac{di_3}{\frac{E}{2R} - \frac{5}{2}i_3} = \frac{R}{L} \int_0^t dt$$

$$\frac{-2}{5} \ln \left(\frac{\frac{E}{2R} - \frac{5}{2}i_3}{\frac{E}{2R}} \right) = \frac{Rt}{2}$$

$$\frac{5}{2}i_3 = \frac{E}{2R} \left(1 - e^{-\frac{5Rt}{2L}} \right)$$

$$i_3 = \frac{E}{5R} \left(1 - e^{-\frac{5Rt}{2L}} \right)$$

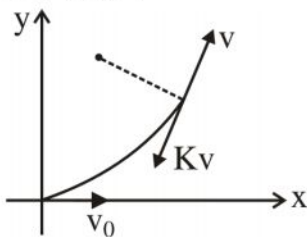
$$i_2 = \frac{E}{2R} - \frac{i_3}{2} = \frac{E}{2R} - \frac{E}{10R} + \frac{E}{10R} e^{-\frac{5Rt}{2L}}$$

$$= \frac{E}{10R} \left(4 + e^{-\frac{5Rt}{2L}} \right)$$

2. Ans (A,B,C,D)

Under normal conditions total energy, potential energy and kinetic energy in ground state and first excited state are -13.6 eV, -27.2 eV, 13.6 eV, -3.4 eV, -6.8 eV and 3.4 eV respectively. If potential energy in ground state is taken to be zero, then kinetic energy will remain unchanged but potential and total energies are increased by 27.2 eV. Therefore, the new values are 13.6 eV, 0 , 13.6 eV, 23.8 eV, 20.4 eV and 3.4 eV respectively.

3. Ans (A,C)



$$r = \frac{mv}{qB}, \omega = \frac{d\theta}{dt} = \frac{qB}{m}$$

For $\theta = 2\pi$, $\vec{v}$ is parallel to +ve x-axis.

$$\Delta t = \frac{\Delta\theta}{\omega} = \frac{2\pi m}{qB}$$

along tangential direction.

$$a_t = \frac{dv}{dt} = -\frac{Kv}{m}$$

$$\int_{v_0}^v \frac{dv}{v} = -\frac{K}{m} \int_0^t dt$$

$$v = v_0 e^{-\frac{K}{m}t}$$

$$\text{distance, } S = \int_0^{\Delta t} v dt$$

$$S = -\frac{v_0}{K/m} \left[e^{-\frac{K}{m}t} \right]_0^{\frac{2\pi m}{qB}}$$

$$S = \frac{mv_0}{K} \left[1 - e^{-\frac{2\pi K}{qB}} \right]$$

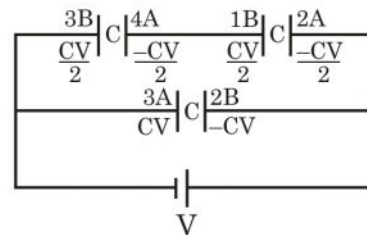
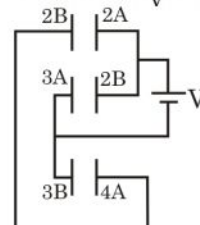
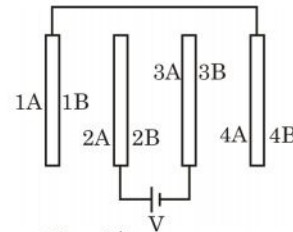
4. Ans (A,C)

$$E_I = \frac{GMx}{R^3}; E_{II} = \frac{GM}{R^3} \frac{R}{2}$$

$$\frac{F_I}{F_{II}} = \frac{2x}{R}$$

$$T_I = \frac{2\pi}{4} \sqrt{\frac{R^3}{GM}}; T_{II} = 2\sqrt{\frac{R^3}{GM}}$$

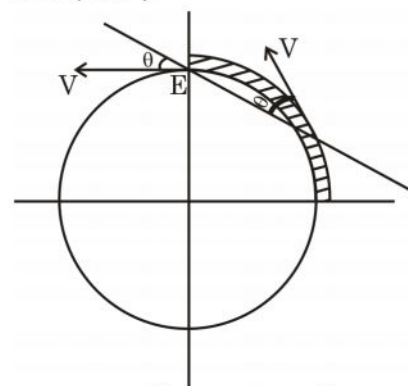
5. Ans (A,B)



$$C_{eq} = \frac{3}{2}C = \frac{3\epsilon_0 A}{2d}$$

$$Q = C_{eq}V = \left(\frac{3\epsilon_0 A}{2d} \right) V$$

6. Ans (C,D)



$$f_{\text{observed}} = f \left[\frac{v_s + v \cos \theta}{v_s + v \cos \theta} \right] = f$$

$$\lambda_{\text{observed}} = \left(\frac{v_s + v \cos \theta}{f} \right)$$

7. Ans (B,C,D)

Momentum along Y-axis will remain conserved

$$\text{So, } mu \cos \theta = 2m V_y$$

$$\Rightarrow V_y = \frac{4u}{10} \quad \dots (i)$$

Conserving angular momentum about Y-axis

during collision

$$mu \sin \theta \ell = \left(\frac{mv^2}{3} + m\ell^2 \right) \omega$$

$$\Rightarrow \omega = \frac{3u \sin \theta}{4\ell} = \frac{9u}{20\ell} \quad \dots (ii)$$

Velocity of particle just after impact is

$$\mathbf{V}_p = V_p \hat{j} + \ell \omega (-\hat{i}) = -\frac{9u}{20} \hat{i} + \frac{4u}{10} \hat{j}$$

for completing full rotation about Y-axis

$$W_g = \Delta k$$

$$\Rightarrow -mg(2\ell) - mg\ell = 0 - \frac{1}{2} \left(\frac{4}{3} m\ell^2 \right) \omega^2$$

$$\Rightarrow \omega_{\min} = \sqrt{\frac{9g}{2\ell}} \Rightarrow u_{\min} = \frac{20\ell}{9} \sqrt{\frac{9g}{2\ell}}$$

$$10\sqrt{\frac{2}{9}g\ell} = \frac{10}{3}\sqrt{2g\ell}$$

PART-1 : PHYSICS

SECTION-I (ii)

9. Ans (C)

$$I = \frac{1}{2} E_0 E_0^2 \cdot C$$

$$570 = \frac{1}{2} E_0 \times \frac{1}{4} \mu_0^2 C^2 J_{\max}^2 \cdot C$$

$$= \frac{1}{8} J_{\max}^2 \cdot C \mu_0$$

$$J_{\max}^2 = \frac{8 \times 570}{3 \times 10 \times 4\pi \times 10^{-7}} = \frac{2 \times 19}{\pi}$$

$$J_{\max} = \sqrt{\frac{38}{\pi}} = 3.48$$

11. Ans (D)

$$p_1 = 1.0 \text{ atm}, V_1 = 1.0L, T_1 = 122 \text{ K}$$

$$p_2 = 1.0 \text{ atm}, V_2 = 2.0L, T_2 = 244 \text{ K}, K = 2T_1$$

$$p_3 = 2.0 \text{ atm}, V_3 = 2.0L, T_3 = 444 \text{ K}, K = 4T_1$$

$$p_4 = 0.0157 \text{ atm}, V_4 = 64L, T_4 = 122 \text{ K}, K = T_1$$

For $1 \rightarrow 2$,

$$Q_1 = nC_p \Delta T = \frac{7}{2} nR (T_2 - T_1)$$

$$= \frac{7}{2} \times n \times R (T_1)$$

For $2 \rightarrow 3$,

$$Q_2 = nC_v \Delta T = \frac{5}{2} nR (T_3 - T_2)$$

$$= \frac{5}{2} \times n \times R (2T_1) = 5nRT_1$$

For $3 \rightarrow 4$,

$$Q_3 = 0$$

For $4 \rightarrow 1$,

$$Q_4 = nRT \ln \frac{V_2}{V_1}$$

$$= nRT_1 \ln \frac{1}{64} = -nRT_1 \ln 64$$

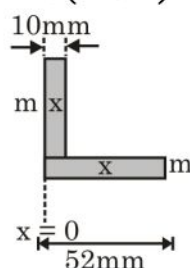
$$\eta = 1 - \frac{Q_-}{Q_+} = 1 - \frac{nRT_1 \ln 64}{\frac{7}{2} nRT_1 + 5nRT_1}$$

$$= 1 - \frac{6 \ln 2}{\frac{17}{2}} = 1 - \frac{12}{17} \ln 2$$

PART-1 : PHYSICS

SECTION-II

1. Ans (21.00)



$$\text{COM of the two dominoes is } x_0 = \frac{m(5) + m(26)}{2m}$$

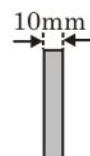
$$x_0 = 15.5 \text{ mm}$$

For system to remain in equilibrium the COM of upper two dominoes must be over the thickness of third domino.

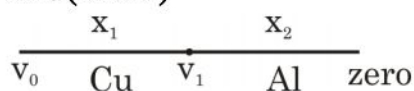
$$x < x_0 < x + 10$$

$$\text{or } x < 15.5 \text{ mm}$$

$$\text{and } 15.5 < x + 10 \Rightarrow x > 5.5 \text{ mm}$$



2. Ans (22.50)



The resistive force is proportional to velocity

$$ma = -kv$$

$$\Rightarrow \frac{mdv}{dt} + kv = 0$$

$$\Rightarrow \frac{d}{dt}(mv + kx) = 0$$

$$\Rightarrow \frac{d}{dt} \left(\frac{m}{k} v + x \right) = 0$$

$$\Rightarrow \frac{m}{k} v + x = C \text{ (constant)(1)}$$

If initial velocity is v_0 and maximum stopping

distance is $x_{\max}$

$$C = \frac{mv_0}{k} = x_{\max} \text{(2)}$$

$$\Rightarrow x_{\max} = \mu v_0 \text{(3)}$$

where $\mu = \frac{m}{k}$ is the characteristic of the metal

plate causing the breaking.

Hence $\mu_{Al}v_0 = 30 \text{ cm}$ and $\mu_{Cu}v_0 = 20 \text{ cm}$

$$\frac{\mu_{Al}}{\mu_{Cu}} = \frac{3}{2}$$

Now, $\mu v + x = C = \mu v_0$

$$\Rightarrow \mu v_1 + 15 = \mu v_0$$

$$v_1 = v_0 - \frac{15}{\mu_{Cu}}$$

$$\text{and } x_2 = \mu_{Al}v_1 = \mu_{Al} \left(v_0 - \frac{15}{\mu_{Cu}} \right)$$

$$\text{and } x_2 = \mu_{Al}v_1 = \mu_{Al} \left(v_0 - \frac{15}{\mu_{Cu}} \right)$$

$$\Rightarrow x_2 = \mu_{Al}v_0 - \frac{\mu_{Al}}{\mu_{Cu}} \times 15$$

$$= 30 - \frac{3}{2} \times 15 = 7.5 \text{ cm}$$

$$x_1 + x_2 = 15 + 7.5 = 22.5 \text{ cm}$$

3. Ans (5.00)

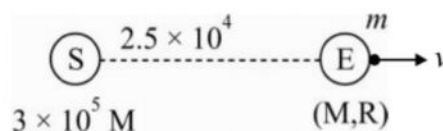
$$\Delta = 70\alpha$$

$$2d \sin \alpha + 2d \sin \Psi = 70 \alpha$$

$$10 \times 10^{-3} \times \frac{10^{-3}}{1} + 10 \times 10^{-3} \cdot \frac{y}{2} = 70\alpha$$

$$y = 5 \text{ mm}$$

4. Ans (40.35 to 40.45)



$$\text{For earth, } V_e = \sqrt{\frac{2GM}{R}}$$

For Sun + Earth,

$$\text{Potential energy} = \left(-\frac{GM}{R} - \frac{3 \times 10^5}{2.5 \times 10^4} \frac{GM}{R} \right) m$$

$$= \frac{GM}{R} (1 + 12) = \frac{13GMm}{R}$$

$$\therefore \frac{1}{2}mv^2 = \frac{13GMm}{R}$$

$$v = \sqrt{\frac{2GM}{R}} \cdot \sqrt{13} = \sqrt{13}v_e$$

$$\therefore v_e = 40.4 \text{ km/s}$$

5. Ans (2.00)

$$\frac{dP}{dy} = -\rho g$$

$$dP = -\rho g dy$$

$$\left[P = \frac{M}{V M_0} RT = \frac{\rho RT}{M_0} \Rightarrow \rho = \frac{P M_0}{RT} \right]$$

$$dP = -\frac{P M_0}{RT} dy$$

$$dP = -\frac{P M_0}{R} \frac{dy}{(T_0 - \alpha y)}$$

$$\int_{P_0}^P \frac{dP}{P} = -\frac{M_0}{R} \int_0^y \left(\frac{dy}{T_0 - \alpha y} \right)$$

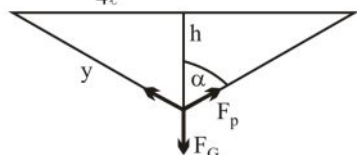
$$\Rightarrow P = P_0 \left(1 - \frac{\alpha y}{T_0} \right)^{\frac{M_0 g}{\alpha R}}$$

6. Ans (9.60 to 9.70)

Since the transverse fibers do not affect the bending of the web, we can only deal with the elongation of the radial fibers.

If a part of the fiber has a length ℓ and a stiffness k , then it is extended by $\Delta x = F/k$ by a force of magnitude F . The radial fiber consists of 4 such parts, therefore it is lengthened by the force $F\Delta X = 4\Delta x$. The total stiffness of the radial fiber will then be (similarly to four springs suspended in series)

$$K = \frac{k}{4}$$



In the middle of the web at point O, the spider acts on the web with its weight, which is in balance with the six elastic forces F_p

$$mg = 6F_p \cos \alpha,$$

$$F_p = K(y - 4\ell) = \frac{k}{4} \left(\sqrt{(4\ell)^2 + h^2} - 4\ell \right)$$

$$\cos \alpha = \frac{h}{y} = \frac{h}{\sqrt{(4\ell)^2 + h^2}},$$

$$mg = 6 \frac{k}{4} \left(\sqrt{(4\ell)^2 + h^2} - 4\ell \right) \frac{h}{\sqrt{(4\ell)^2 + h^2}}$$

$$k = \frac{2mg\sqrt{(4\ell)^2 + h^2}}{3h \left(\sqrt{(4\ell)^2 + h^2} - 4\ell \right)}$$

$$k = \frac{2mg}{3h} \times \frac{4\ell}{4\ell \left(\frac{\sqrt{1+h^2}}{(4\ell)^2} - 1 \right)}$$

$$= \frac{2mg}{3h} \frac{1}{\left(1 + \frac{1}{2} \frac{h^2}{(4\ell)^2} - 1 \right)} = \frac{2mg}{3h} \times \frac{2 \times 16\ell^2}{h^2}$$

$$= \frac{64}{3} \frac{mg\ell}{h^3}$$

$$= \frac{64}{3} \times \frac{0.01 \times 10 \times 0.1 \times 0.1}{125 \times 10^{-9}} = 0.17 \times 10^6 \text{ N/m}$$

The modulus of elasticity in tension will then be according to Hooke's law

$$E = \frac{\sigma}{\epsilon} = \frac{F_p}{S \frac{\Delta x}{\ell}} = \frac{4k\ell \Delta x}{\pi d^2 \Delta x} = \frac{4k\ell}{\pi d^2}$$

$$E = \frac{4 \times 0.17 \times 10^6 \times 0.1}{\frac{22}{7} \times (0.15 \times 0.15) \times 10^{-12}} = \frac{0.68 \times 10^5}{0.07065 \times 10^{-12}} = 9.62 \times 10^{17} \text{ Pa}$$

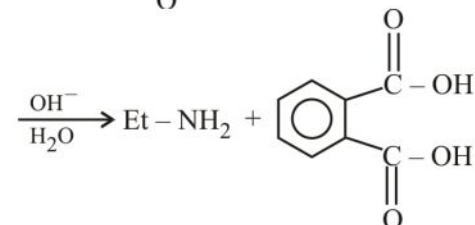
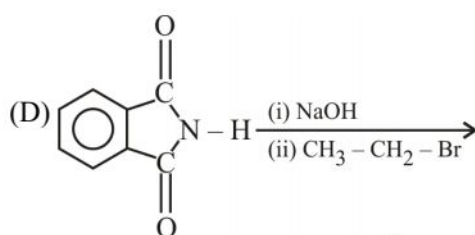
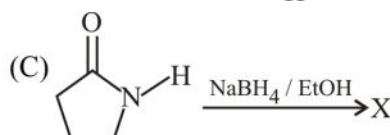
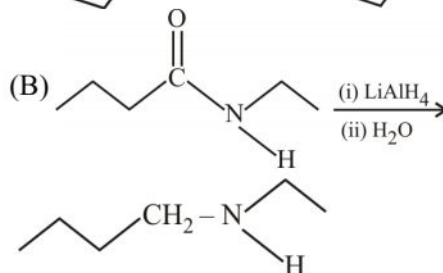
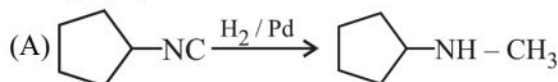
PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (B,C)

Coordination entity	Wavelength of light absorbed (nm)	
$[\text{CoCl}(\text{NH}_3)_5]^{2+}$	(P)	535
$[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$	(Q)	500
$[\text{Co}(\text{NH}_3)_6]^{3+}$	(R)	475
$[\text{Co}(\text{CN})_6]^{3-}$	(S)	310
$[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$	(T)	600

3. Ans (A,B)



6. **Ans (B,C)**

$$P_A = 235 y_A - 125 x_B y_A$$

$$= y_A [235 - 125 (1 - x_A)]$$

$$= y_A [235 - 125 + 125 x_A]$$

$$P_A = y_A [110 + 125 x_A]$$

$$P_A = P_T \times y_A$$

$$\Rightarrow P_T = (P_A^\circ - P_B^\circ) x_A + P_B^\circ$$

$$\therefore P_B^\circ = 110$$

$$P_A^\circ - P_B^\circ = 125$$

$$P_A^\circ = 235.$$

7. **Ans (B,C)**

(A) Coagulation value.

(B) Coagulation power $\propto \frac{1}{\text{Coagulation value}}$

(C) Hardy Schultz rule

(D) Definition of protective colloid.

PART-2 : CHEMISTRY

SECTION-I (ii)

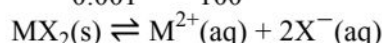
8. **Ans (D)**

$$\Delta G = -nFE_{\text{cell}} = \frac{-2 \times 96500 \times 0.059}{1000} \text{ KJ mol}^{-1}.$$

9. **Ans (B)**

$$0.059 = \frac{-0.059}{2} \log \frac{[M^{2+}]_A}{0.001}$$

$$\text{or, } \frac{[M^{2+}]_A}{0.001} = \frac{1}{100} \text{ or, } [M^{2+}]_A = 10^{-5} = s(\text{say})$$



$$s \quad 2s$$

$$K_{sp} = 4s^3 = 4 \times 10^{-15}$$

10. **Ans (A)**

$$n = 2 \rightarrow \begin{matrix} \ell = 0 & 1 & 2 & 3 \\ s & p & d & f \end{matrix}$$

No change in the value of m

$\therefore$ total orbitals associated to 2nd shell = 16

$$\text{The value of } s = +\frac{1}{4}, +\frac{1}{2}, -\frac{1}{2}, -\frac{1}{4}$$

$\therefore$ total no of e⁻ associated to 2nd shell = 16 $\times$ 4 = 64

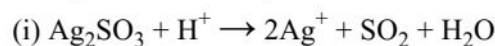
11. **Ans (B)**

p-orbital having 4e⁻ (according to value of 's')

PART-2 : CHEMISTRY

SECTION-II

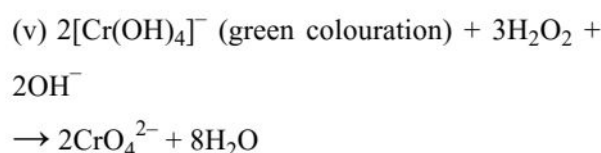
2. **Ans (13.66 to 13.67)**



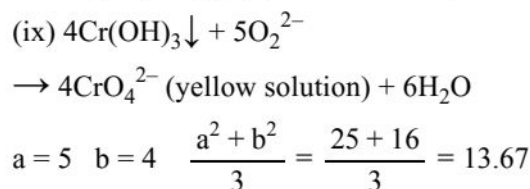
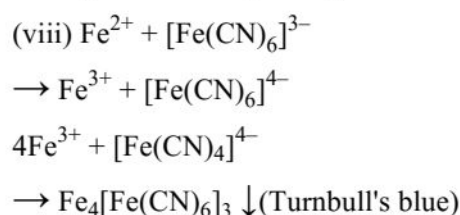
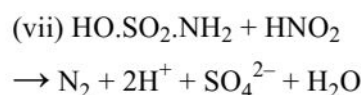
(ii) FeSO₄ solution absorbs NO and becomes brown because of the formation of [Fe(NO)]²⁺SO₄²⁻.

(iii) Being covalent do not dissociate

(iv) Ammonium hydroxide in presence of ammonium chloride is group reagent of IIIrd group.



(vi) Red colour solution or precipitate is produced when reagent reacts in alkaline solution.



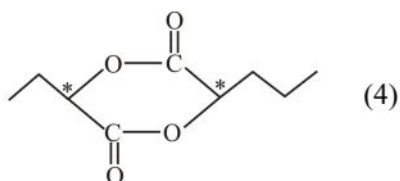
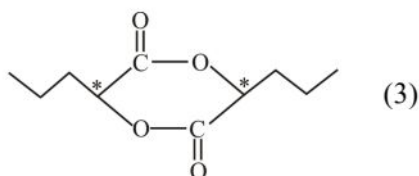
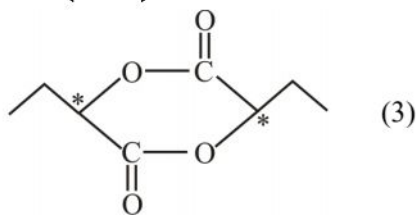
3. **Ans (9.00)**

$$x = 5, y = 4$$

Tollen's test : (i), (ii), (iv), (vi), (vii)

Fehling test : (i), (ii), (iv), (vi)

4. Ans (5.00)



5. Ans (-530.00)

$dG = V.dP - S.dT$; at const. volume

$$\Delta G = V.\Delta P - (10 + 10^{-2} T).dT \quad \frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow$$

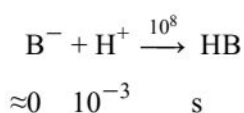
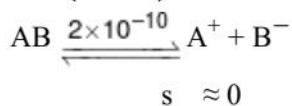
$$P_2 = 1 \times 400/300$$

$$\Delta G = 24.6 \times \left(\frac{4}{3} - 1\right) \times 100 - \left[10 \times 100 + 10^{-2} \times \left(\frac{T_2^2}{2} - \frac{T_1^2}{2}\right)\right]$$

$$\Delta G = 24.6 \times \frac{1}{3} \times 100 - [1000 + 10^{-2} \times \left(\frac{160000}{2} - \frac{90000}{2}\right)]$$

$$= 24.6 \times \frac{1}{3} \times 100 - 1000 - 350 = -530 \text{ J}$$

6. Ans (20.00)



$$2 \times 10^{-10} = s \times [B^-] \quad \dots\dots (1)$$

$$10^8 = \frac{s}{[B^-] \times 10^{-3}} \Rightarrow 10^8 = \frac{s^2}{2 \times 10^{-10} \times 10^{-3}}$$

$$\Rightarrow s^2 = 2 \times 10^{-5} \Rightarrow s = \sqrt{20} \times 10^{-3}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,C)

$$\text{Required Area} = 2 \int_{-3/2}^0 2 \left(x + \frac{3}{2}\right) dx = 2\sqrt{3}$$

2. Ans (A,D)

$$A = \int_0^1 [-\ln y] - (e^y - e) dy = 2$$

3. Ans (A,B,C)

$$3x^2 \frac{dx}{dy} = \frac{x^3}{x^3 - y} \quad x^3 = t$$

$$\frac{dt}{dy} = \frac{t}{t - y} \Rightarrow t dt = y dt + x dy$$

$$\frac{t^2}{2} = t \cdot y + \frac{c}{2}$$

$$\Rightarrow x^6 = 2x^3 y + c$$

4. Ans (A,B,C)

$$\frac{n^4 + 3n^2 + 10n + 10}{2^n(n^4 + 4)}$$

$$= \frac{1}{2^n} + \frac{3n^2 + 10n + 6}{2^n(n^2 + 2n + 2)(n^2 - 2n + 2)}$$

$$= \frac{1}{2^n} + \frac{4n^2 + 8n + 8 - n^2 + 2n - 2}{2^n(n^2 + 2n + 2)(n^2 - 2n + 2)}$$

$$= \frac{1}{2^n} + \frac{1}{2^{n-2}(n^2 - 2n + 2)} - \frac{1}{2^{n-2}(n^2 + 2n + 2)}$$

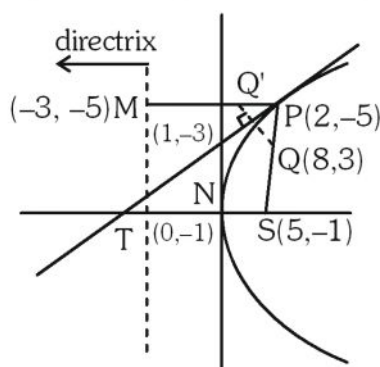
$$\therefore \sum_{n=2}^{\infty} \frac{n^4 + 3n^2 + 10n + 10}{2^n(n^4 + 4)}$$

$$= \sum_{n=2}^{\infty} \frac{1}{2^n} + \sum_{n=2}^{\infty} \left(\frac{1}{2^{n-2}(n^2 - 2n + 2)} - \frac{1}{2^{n-2}(n^2 + 2n + 2)} \right)$$

$$= \frac{1}{2} + \frac{6}{10} = \frac{11}{10}$$

5. Ans (A,C)

Equation of tangent at P is $2x + y + 1 = 0$



N will be the mid point of PT

$\Rightarrow$ T is (0, -1)

Image of Q(8, 3) w.r.t. to tangent at P lies on PM.

Image Q' of Q(8, 3) w.r.t. $2x + y + 1 = 0$

$$\frac{x-8}{2} = \frac{y-3}{1} = \frac{-2[16+3+1]}{5}$$

$\Rightarrow x = -8, y = -5$

Q'(-8, -5)

$\Rightarrow$ PM will be $y = -5$

$\Rightarrow$ axis of parabola will be $y = -1$

PQ is a focal chord of parabola equation of PQ is

$4x - 3y - 23 = 0$ focus will be point of intersection

axis & PS

$\Rightarrow$ focus S is (5, -1) Point M lies on directrix of

parabola N will be mid point of M & S

$\Rightarrow$ M is (-3, -5)

$\Rightarrow$ directrix is $x = -3$

Normal at other end of focal chord at P will be parallel to tangent at P.

Circle described on focal radii as diameter touches tangent at vertex of parabola.

6. Ans (A,B,D)

Let $(x^2 - 4x + 6) = t$

$$f(t) = t^2 - (k+2)t + k + 5 = 0$$

Apply conditions of location of roots on equation $f(t) = 0$.

for exactly 4 real solution $k \in (4, 5)$

for exactly 3 real solutions $k = 5$

for exactly are real solution $k \in \phi$

for no real solutions $k \in (-\infty, 4)$

7. Ans (A,B,C)

$$a \left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots \right) + b \left(x - \frac{x^3}{3} + \frac{x^5}{5} + \dots \right) + c \left(x + \frac{x^3}{6} + \frac{9x^5}{5!} + \dots \right)$$

$$\lim_{x \rightarrow 0} \frac{\dots}{x^5} = 3$$

$$(a+b+c)x + \left(\frac{a}{3} - \frac{b}{3} + \frac{c}{6} \right) x^3$$

$$+ \left(\frac{2a}{15} + \frac{b}{5} + \frac{9c}{5!} \right) x^5 + \dots$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\dots}{x^5} = 3$$

$$a + b + c = \dots(1)$$

$$2a - 2b + c = 0 \dots(2)$$

$$\frac{2a}{15} + \frac{b}{5} + \frac{3c}{40} = 3 \dots(3)$$

$$\Rightarrow a = 30, b = 10, c = -40$$

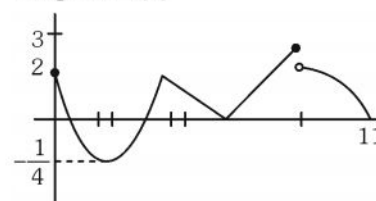
$$2a + b + c = 60 + 10 - 40 = 30$$

PART-3 : MATHEMATICS

SECTION-I (ii)

8. Ans (A)

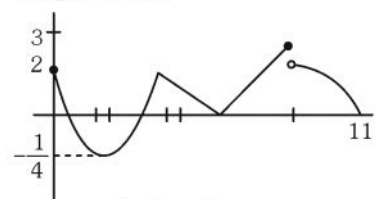
Graph of $f(x)$



$$\text{Range : } \left[-\frac{1}{4}, 3 \right]$$

9. Ans (B)

Graph of $f(x)$



$$\text{Range : } \left[-\frac{1}{4}, 3 \right]$$

10. Ans (B)

$$(A) \frac{d}{dx} f(x) = \frac{e^{\sin x}}{x}$$

$$f(x) = \int \frac{e^{\sin x}}{x} dx$$

$$\Rightarrow \int_1^4 \frac{2e^{\sin x^2}}{x} dx = \int_1^4 \frac{e^{\sin(x^2)}}{x^2} d(x^2)$$

$$x^2 = t$$

$$= [f(t)]_1^{16}$$

$$= f(16) - f(1)$$

$$\Rightarrow \boxed{k = 16}$$

$$(B) xf'(x) = e^{\sin x}$$

$$xf''(x) + f'(x) = e^{\sin x} \cdot \sin x$$

$$= g(x)$$

11. Ans (A)

$$(A) \frac{d}{dx} f(x) = \frac{e^{\sin x}}{x}$$

$$f(x) = \int \frac{e^{\sin x}}{x} dx$$

$$\Rightarrow \int_1^4 \frac{2e^{\sin x^2}}{x} dx = \int_1^4 \frac{e^{\sin(x^2)}}{x^2} d(x^2)$$

$$x^2 = t$$

$$= [f(t)]_1^{16}$$

$$= f(16) - f(1)$$

$$\Rightarrow \boxed{k = 16}$$

$$(B) xf'(x) = e^{\sin x}$$

$$xf''(x) + f'(x) = e^{\sin x} \cdot \sin x$$

$$= g(x)$$

$$(C) I = \int_0^{\pi/2} x \cos x \cdot \frac{e^{\sin x}}{x} dx$$

$$\sin x = t$$

$$\cos x dx = dt$$

$$= \int_0^1 e^t dt = e^1 - 1$$

$$\lambda = \frac{\pi}{2}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.75)

If the one of the digit first or last is zero then number of numbers is $8 \times 7 = 56$.

If none of the first and last digit is zero then number of numbers are $2 \times 8 \times 7 \times 7$

Hence total numbers = $8 \times 7 + 2 \times 8 \times 7 \times 7 = 840$

2. Ans (1.51)

$$a = -\sqrt{2}, b = \sqrt{2}$$

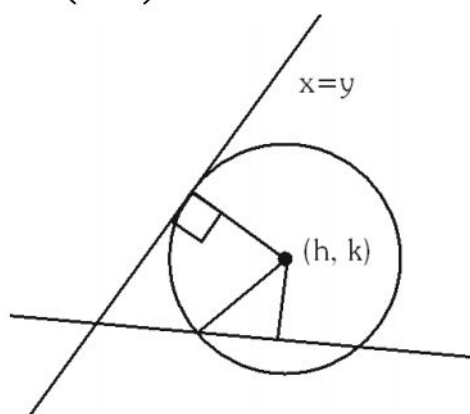
$$\text{area} = - \int_{-\sqrt{2}}^{\sqrt{2}} (x^4 - 2x^2) dx$$

$$= -2 \int_0^{\sqrt{2}} (x^4 - 2x^2) dx$$

$$= 2 \int_0^{\sqrt{2}} (2x^2 - x^4) dx$$

$$\therefore \text{Area} = \frac{16\sqrt{2}}{15} = 1.51$$

3. Ans (2.82)



$$\left(\frac{h-k}{\sqrt{2}} \right)^2 = \left(\frac{h+k}{\sqrt{2}} \right)^2 + 2$$

$$\therefore xy = -1$$

Hence locus is rectangular hyperbola with length of latus rectum = $2\sqrt{2}$

4. Ans (1.80)

$$\lim_{n \rightarrow \infty} n \sum_{r=1}^n \frac{1}{3n^2 + 8rn + 4r^2}$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \left(\frac{1}{3 + 8\left(\frac{r}{n}\right) + 4\left(\frac{r}{n}\right)^2} \right)$$

$$= \frac{1}{4} \int_0^1 \frac{dx}{x^2 + 2x + 1 - \frac{1}{4}} = \frac{1}{4} \int_0^1 \frac{dx}{(x+1)^2 - \left(\frac{1}{2}\right)^2}$$

$$\text{Thus } \frac{1}{4} \ell n \left| \frac{x+1 - \frac{1}{2}}{x+1 + \frac{1}{2}} \right| = \frac{1}{4} \ell n \frac{9}{5}$$

5. Ans (5.50)

$$\sigma_{12}^2 = \frac{N_1 \sigma_1^2 + N_2 \sigma_2^2 + N_1 d_1^2 + N_2 d_2^2}{N_1 + N_2}$$

Here σ_{12}^2 = combined variance

$$N_1 = N_2 = 5, \sigma_1^2 = 4, \sigma_2^2 = 5, d_1 = -1, d_2 = 1$$

6. Ans (2.00)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$AA^T = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{bmatrix}$$

$$\therefore a^2 + b^2 = 1, c^2 + d^2 = 1, ac + bd = 0$$

$$\text{Let } a = \cos \theta, b = \sin \theta, c = \cos \phi, d = \sin \phi$$

$$\therefore \cos \theta \cos \phi + \sin \theta \sin \phi = 0, \therefore \theta - \phi = \frac{\pi}{2}$$

$$\Rightarrow |a| + |b| + |c| + |d| = 2 (|\cos \theta| + |\sin \theta|)$$

$$\text{since } |\cos \theta| + |\sin \theta| \leq \sqrt{2}$$

$$\text{greatest value of } 2 (|a| + |b|) = 2\sqrt{2}$$